

NAME:

MATH 316 – HW 6

Due: Tuesday Oct 6, 2026

HOMEWORK DAY 21 – *Laplace Transform – Definition §3.1*

1. p144: 1.

2. Using the definition, show that $\mathcal{L}[t^2] = \frac{2}{s^3}$. You may use the transform $\mathcal{L}[t]$ derived in class.

Use the formula for $\mathcal{L}[e^{at}]$ derived in class for the following problems

3. p145: 3.

4. p145: 4.